Notes: Hartzmark & Shue (JF, 2018)
A Tough Act to Follow: Contrast Effects in Financial Markets

Contents

1 Big picture

- **Question.** Can contrast effects distort prices in sophisticated financial markets? A contrast effect means that a previously observed signal inversely biases the perceived quality of the next signal.
- **Setting.** Quarterly earnings announcements by U.S. public firms. The paper studies whether the market reaction to a firm's earnings news today depends on the earnings surprises released by large firms on the previous trading day.
- **Core idea.** Today's earnings surprise should be evaluated by its own information content. But if yesterday's salient earnings news was very good, today's news may look less impressive; if yesterday's news was bad, today's news may look more impressive.
- **Sample.** The main analyst-based sample contains 75,897 earnings announcements from 1984 to 2013. The key previous-day benchmark, $Surprise_{t-1}$, is the value-weighted earnings surprise of large firms that announced on the previous trading day.
- **Main result.** Holding the firm's own earnings surprise fixed, the return reaction to today's earnings announcement is negatively related to yesterday's salient surprise. The baseline close-to-close coefficient is -0.924 and is significant at the 1% level.
- **Magnitude.** Moving from the lowest to the highest decile of yesterday's salient surprise is associated with roughly a 53 bp lower return reaction to today's earnings announcement.
- **Mechanism.** The effect is mostly directional. Good previous news shifts the whole response curve downward; the interaction with today's own surprise is small and insignificant.
- **Mispricing evidence.** The initial distortion reverses after the announcement, especially over days $t + 26$ to $t + 50$, consistent with an error that is later corrected.
- **Trading implication.** A simple strategy that buys firms announcing today after bad previous-day surprises and shorts them after good previous-day surprises earns significant abnormal returns in the short window $[t, t + 1]$.
- **Identification advantage.** Because stock prices are continuously traded, the paper can distinguish perceptual errors from expectational errors. Prices do not move on day $t - 1$ as if yesterday's surprise predicts today's firm; the distortion appears when today's earnings news is interpreted.
- **Economic takeaway.** Investors do not only react to the absolute content of news. They also evaluate news relative to recently observed salient news, and this relative perception can move prices even in liquid equity markets.

2 Data and measurement

2.1 Why earnings announcements are a useful laboratory

- Earnings announcements are recurring, information-rich, and widely followed.
- Actual earnings can be compared with analyst forecasts to construct a clear measure of surprise.
- Announcement dates are usually scheduled in advance, so whether a firm announces after good or bad news from other large firms is plausibly not chosen based on the firm's own fundamentals.
- Return reactions around announcements allow the authors to measure market perception of news.

Interpretation. If contrast effects exist here, they affect equilibrium prices rather than only survey responses or lab judgments. This is why the paper is important for behavioral asset pricing.

2.2 Main datasets

- **I/B/E/S.** Analyst forecasts, forecast dates, actual earnings, and earnings announcement dates.

- **CRSP.** Daily stock returns.
- **Compustat.** Firm characteristics such as market capitalization and accounting variables.
- **Kenneth French Data Library.** Risk factors, size cutoffs, and benchmark portfolios.

2.3 Own earnings surprise

The firm-level earnings surprise is

$$Surprise_{it} = \frac{Actual\ earnings_{it} - Median\ estimate_{i,[t-15,t-2]}}{Price_{i,t-3}}. \tag{1}$$

The median estimate is the median of each analyst's most recent forecast made from 15 to 2 days before the announcement.

Interpretation. Scaling by $Price_{i,t-3}$ makes surprises comparable across firms with different share prices. The exclusion of days t and $t-1$ ensures that the forecast information is observed before the previous-day salient announcement.

2.4 Previous-day salient surprise

The paper's baseline measure of the previous-day signal is

$$Surprise_{t-1} = \frac{\sum_{i=1}^{N} Mktcap_{i,t-4} \times Surprise_{i,t-1}}{\sum_{i=1}^{N} Mktcap_{i,t-4}}, \tag{2}$$

where the sum is over large firms that announced on the previous trading day. A firm is "large" if its market capitalization exceeds the NYSE 90th percentile cutoff in that month.

Interpretation. Size is used as a proxy for salience. A very large firm's earnings announcement is more likely to be noticed by investors evaluating other firms' earnings the next day.

2.5 Return measure

The main dependent variable is the firm's raw return minus the return of a characteristic-matched portfolio. The matching follows the Daniel-Grinblatt-Titman-Wermers style approach using size, book-to-market, and momentum quintiles. The matched portfolio excludes the announcing firm and firms used to construct $Surprise_{t-1}$.

Interpretation. This adjustment removes common return movements related to standard firm characteristics and prevents mechanical overlap between the previous-day benchmark and today's adjusted return.

2.6 Event windows

- $[t, t+1]$: open-to-open return used when the previous-day return is not included.
- $[t-1, t+1]$: close-to-close return used in the baseline direct test, allowing the market to react from before the current announcement.
- $[t+2, t+50]$ and related windows: long-run reversal tests.

Interpretation. The short window measures the immediate perception of today's news. The longer windows test whether the initial price movement is corrected later.

3 Models and empirical specifications

3.1 Contrast-effects mechanism

Informally, the paper's mechanism is

$$\textit{Good salient news}_{t-1} \Rightarrow \textit{today's news looks worse} \Rightarrow \textit{lower return reaction}_t. \tag{3}$$

The reverse holds after bad salient news on $t-1$.

Interpretation. The key implication is not that investors expect today's firm to be worse after good news yesterday. The implication is that after seeing today's news, investors perceive it through a biased relative comparison.

3.2 Unconditional predictability test

The simple regression is

$$Return_{i,[t,t+1]} = \beta_0 + \beta_1 Surprise_{t-1} + \varepsilon_{it}. \tag{4}$$

Interpretation. Under efficient processing of public information, yesterday's surprise should not predict today's returns. A negative β_1 says that after good previous-day news, firms scheduled to announce today earn lower returns.

3.3 Direct contrast-effects test

The core direct specification controls for the firm's own news:

$$Return_{i,[t-1,t+1]} = \beta_0 + \beta_1 Surprise_{t-1} + Own\ surprise\ bin_{it} + \varepsilon_{it}. \tag{5}$$

The paper uses 20 bins of the firm's own earnings surprise.

Interpretation. This asks whether the same current news is priced differently depending on what investors saw yesterday. The coefficient $\beta_1 < 0$ is the direct contrast-effects estimate.

3.4 Interaction test

The paper also estimates

$$Return_{i,[t-1,t+1]} = \beta_0 + \beta_1 Surprise_{t-1} + \beta_2 Surprise_{t-1} \times OwnSurprise_{it} + controls + \varepsilon_{it}. \tag{6}$$

Interpretation. The estimates show that the distortion does not depend strongly on whether today's surprise is positive or negative. A high previous-day surprise shifts the response curve downward.

3.5 Reversal tests

The paper estimates the baseline relation over later windows:

$$Return_{i,W} = \beta_0 + \beta_1 Surprise_{t-1} + Own\ surprise\ bins + \varepsilon_{it}, \tag{7}$$

where W can be $[t+2, t+25]$, $[t+26, t+50]$, or $[t+51, t+75]$.

Interpretation. If the initial effect is mispricing, β_1 should turn positive in the correction window. That is what the paper finds around $t+26$ to $t+50$.

3.6 Trading strategy

The strategy is:

- go long firms announcing today if $Surprise_{t-1}$ is below the prior-quarter 25th percentile;
- go short firms announcing today if $Surprise_{t-1}$ is above the prior-quarter 75th percentile;
- hold the event-window position over $[t, t+1]$;
- compute abnormal returns using market, SMB, HML, momentum, and short-term reversal factors.

Interpretation. The strategy translates the psychological distortion into a return-predictability test.

4 Identification and empirical design

4.1 What identifies the main effect

- The previous-day signal is public information from other large firms.
- The current firm's own earnings surprise is directly controlled for.
- Announcement timing is typically predetermined, so the sequence of announcements is plausibly unrelated to the current firm's unobserved fundamentals.
- Standard errors are clustered by date, which matters because many firms announce on the same day.

4.2 Why this is not simple information transmission

- $Surprise_{t-1}$ does not reliably predict today's own earnings surprises once time trends are controlled.
- Firms scheduled to announce on day t do not move on day $t-1$ as if yesterday's news is already informative about them.
- The initial return distortion reverses later, which is hard to reconcile with rational processing of value-relevant information.

4.3 Why this is not just risk, liquidity, or capital constraints

- The effect survives controls for market factors and interactions with those factors.
- It is not explained by daily trading volume or bid-ask spreads.
- It survives controls for market returns, VIX, downturn quarters, and a measure of earnings-arbitrage capital.

4.4 Why strategic scheduling is unlikely to drive the effect

The paper compares firms whose announcement date is close to the previous year's same-quarter announcement date with firms that shift their dates more substantially. The effect remains strongest when the date shift is small.

Interpretation. This helps rule out the concern that managers deliberately schedule earnings to exploit previous earnings news.

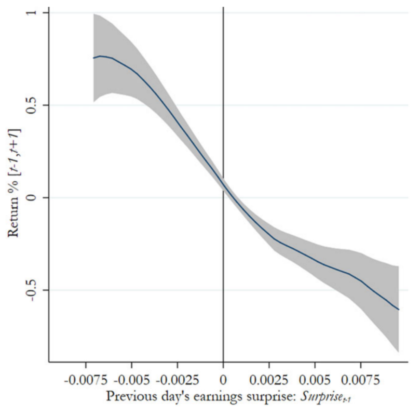
4.5 What the paper cannot fully prove

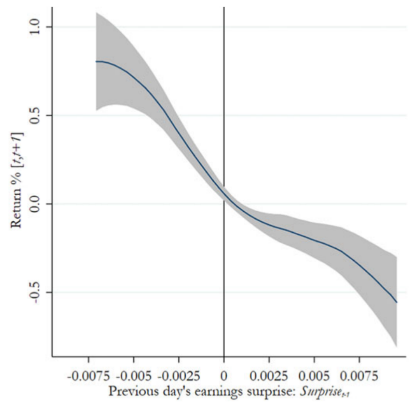
- It cannot observe each investor's true mental comparison set.

- It cannot prove that all investors are biased; prices may move because biased investors are marginal and arbitrage is limited or unaware.
- It cannot rule out every possible information-transmission story, but the alternative would need to combine delayed reaction, negative cross-firm news correlation, and long-run reversal.

5 Tables and figures

5.1 Figures 1 and 2: return reactions to previous-day surprise





Read: Figures 1 and 2 both show a downward-sloping relation between yesterday's salient earnings surprise and today's announcement return. Figure 1 conditions on the firm's own earnings surprise; Figure 2 shows the unconditional relation.

Economic message: Good earnings news yesterday makes today's earnings announcement look worse; bad news yesterday makes today's news look better. This is the visual core of the paper.

5.2 Tables I and II: summary statistics and baseline results

Table I
Summary Statistics

This table presents summary statistics for the main variables used in our analysis using data from 1984 to 2013. The earnings surprise is measured as $(Actual - Forecast)/Price_{t-3}$, where *Forecast* is the median of each analyst's most recent forecast that is released within 15 days of the announcement, excluding t and $t-1$. Returns are the return of a firm minus the return of a portfolio matched on quintiles of market capitalization, book-to-market ratio, and momentum (excluding firms used in the calculation of $Surprise_{t-1}$ and the announcing firm). $Surprise_{t-1}$ is our baseline measure of the salient surprise released by other firms on the previous trading day. It is calculated as the value-weighted earnings surprise of all large firms that announced on the previous trading day. We define a "large" firm as a firm with market capitalization three days before its earnings announcement that exceeds the 90$^{\text{th}}$ percentile cutoff of the NYSE index that month.

	N (1)	Mean (2)	SD (3)	P25 (4)	P50 (5)	P75 (6)
Own Earnings Surprise	75,897	−0.0003	0.0137	−0.0003	0.0002	0.0015
Return [$t-1$, $t+1$]	75,897	0.0018	0.0701	−0.0315	0.0007	0.0351
Return [$t-1$, $t+1$] (open-to-open)	61,640	0.0009	0.0693	−0.0314	0.0004	0.0340
Market Cap$_{t-3}$ [Millions of Dollars]	75,897	7,679	24,100	441	1,491	5,069
Number of Analysts [$t-15$, $t-2$]	75,897	3.722	3.671	1	2	5
$Surprise_{t-1}$	75,897	0.0005	0.0017	0.0000	0.0004	0.0010
Number of large firm surprises [$t-1$]	75,897	7.578	5.800	3	6	12

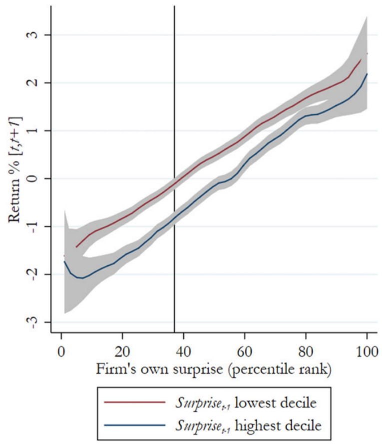
Table II
Baseline Results

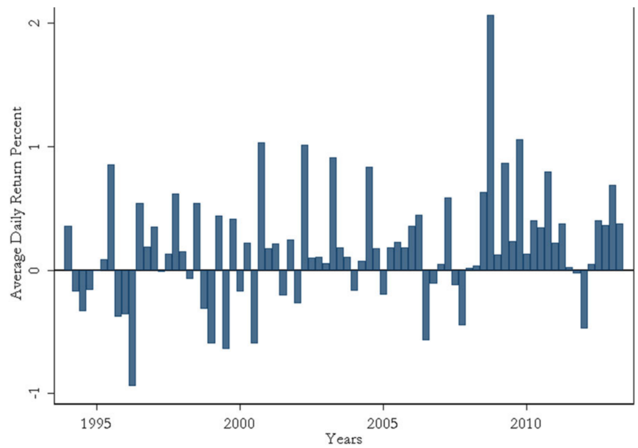
	Open-to-Open Return [$t, t+1$]			Close-to-Close Return [$t-1, t+1$]		
	(1)	(2)	(3)	(4)	(5)	(6)
$Surprise_{t-1}$ of largest firm	-0.526^{***} (0.199)			-0.608^{***} (0.178)		
$Surprise_{t-1}$ large firms, EW mean		-0.846^{***} (0.289)			-1.041^{***} (0.254)	
$Surprise_{t-1}$ large firms, VW mean			-0.780^{***} (0.240)			-0.924^{***} (0.229)
Own $surprise_{it}$ controls	No	No	No	Yes	Yes	Yes
R^2	0.000584	0.000753	0.000841	0.0596	0.0600	0.0600
Observations	61,640	61,640	61,640	75,897	75,897	75,897

Read: Table I shows the main sample: 75,897 announcements, a median own earnings surprise close to zero, and a median of six large-firm surprises used to construct $Surprise_{t-1}$. Table II shows the baseline negative effect. The preferred value-weighted, close-to-close specification gives -0.924^{***}.

Economic message: The data are large and the effect is economically meaningful. The return reaction depends not only on today's news, but also on yesterday's salient comparison news.

5.3 Figures 3 and 4: response-curve shift and trading-strategy returns





Read: Figure 3 splits firms by whether the previous-day surprise was in the lowest or highest decile. The high-$Surprise_{t-1}$ curve is shifted downward across the range of the firm's own surprise. Figure 4 shows that the simple contrast strategy earns positive returns over many quarters, with a large spike in 2008.

Economic message: Contrast effects are mostly a vertical shift in perceived news quality. The pricing error is strong enough to create tradable return predictability.

5.4 Tables III and IV: interaction effects and long-run reversals

Table III

Potential Interaction Effects

This table examines whether contrast effects are related to an interaction between $Surprise_{t-1}$ and the firm's announced surprise on day t. Column (1) measures the surprise today using levels, column (2) measures it using 20 equally sized bins, and column (3) uses quintiles. For brevity, we report only the interaction effects, but all direct effects are included in the regressions. All other variables and weights are as defined in Table II. Standard errors are clustered by date and reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Return $[t-1, t+1]$		
	(1)	(2)	(3)
$Surprise_{t-1}$	-0.703^{***}	-1.468^{***}	-1.442^{**}
	(0.227)	(0.529)	(0.722)
$Surprise_{t-1}$ × own surprise	14.56		
	(40.05)		
$Surprise_{t-1}$ × own surprise (20 bins)		0.0581	
		(0.0497)	
$Surprise_{t-1}$ × own surprise quintile 2			0.257
			(0.909)
$Surprise_{t-1}$ × own surprise quintile 3			0.682
			(0.929)
$Surprise_{t-1}$ × own surprise quintile 4			0.844
			(0.831)
$Surprise_{t-1}$ × own surprise quintile 5			0.893
			(1.082)
R^2	0.0123	0.0555	0.0562
Observations	75,897	75,897	75,897

Table IV

Long-Run Reversals

This table examines the relation between $Surprise_{t-1}$ and long-run return reactions. Return windows are as indicated in column headers. All other variables and weights are as defined in Table II. Standard errors are clustered by date and reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Return Window:	$[-1, +1]$ (1)	$[-1, +25]$ (2)	$[-1, +50]$ (3)	$[-1, +75]$ (4)	$[+2, +25]$ (5)	$[+26, +50]$ (6)	$[+51, +75]$ (7)
$Surprise_{t-1}$	-0.924^{***}	-0.860^{*}	0.365	0.100	0.112	1.152^{**}	-0.316
	(0.229)	(0.515)	(0.686)	(0.776)	(0.431)	(0.475)	(0.522)
Own $Surprise_{it}$ controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0600	0.0222	0.0115	0.00862	0.00189	0.00236	0.00270
Observations	75,897	75,428	74,122	73,478	75,428	74,122	73,478

Read: Table III shows that interaction terms between yesterday's surprise and today's own surprise are not statistically important. Table IV shows the initial negative effect reverses, with a significant positive coefficient of 1.152^{**} in the $[t+26, t+50]$ window.

Economic message: The evidence looks like a perceptual distortion rather than correct information. Prices initially overreact in the contrast-biased direction and later correct.

5.5 Tables V and VI: trading strategy and information transmission

Table V
Trading Strategy

	$[t, t+1]$ (1)	$[t+2, t+50]$ (2)	$[t+26, t+50]$ (3)	$[t+51, t+75]$ (4)
Alpha [%]	0.189***	−0.0257**	−0.0407**	−0.0104
	(0.0558)	(0.0127)	(0.0196)	(0.0226)
Mkt	−0.0371	0.0311***	0.0415**	0.0462**
	(0.0461)	(0.0115)	(0.0180)	(0.0206)
SMB	0.0693	0.0238	−0.00250	−0.0381
	(0.0871)	(0.0216)	(0.0333)	(0.0382)
HML	−0.182**	−0.0107	−0.0700**	0.0222
	(0.0825)	(0.0222)	(0.0340)	(0.0390)
UMD	0.000897	0.00143	0.0336	−0.0846***
	(0.0597)	(0.0147)	(0.0227)	(0.0261)
ST Reversal	−0.0909	0.000462***	0.000834***	−0.000517**
	(0.0716)	(0.000149)	(0.000230)	(0.000263)
Observations	1,525	5,064	4,781	4,773

Table VI
Information Transmission

	$Surprise_{it}$		20 $surprise_{it}$ bins		Close-to-Close ret $[t-1]$	Open-to-Open ret $[t-1]$
	(1)	(2)	(3)	(4)	(5)	(6)
$Surprise_{t-1}$	0.153***	0.00660	134.2***	−30.33	0.0366	0.117
	(0.0565)	(0.0563)	(32.86)	(27.47)	(0.135)	(0.161)
Year-month FE	No	Yes	No	Yes	No	No
R^2	0.00194	0.0325	0.00286	0.0649	0.0000122	0.000125
Observations	75,897	75,897	75,897	75,897	75,897	61,640

Read: Table V reports a 0.189*** daily alpha for the immediate $[t, t+1]$ trading strategy. Reversal strategy returns are negative over later windows, consistent with correction of the original mispricing. Table VI shows that $Surprise_{t-1}$ does not predict today's earnings once year-month fixed effects are included and does not move today's announcers on day $t-1$.

Economic message: The market is not simply incorporating real cross-firm information. The return predictability appears when today's news is perceived and interpreted.

5.6 Table VII: risk, trading frictions, and limited capital

Table VII
Risk, Trading Frictions, and Limited Capital

Panel A: Risk and Liquidity

	Return $[t-1, t+1]$ (1)	Raw ret $[t-1, t+1]$ (2)	Log(volume) (3)	Log(bid-ask) (4)
$Surprise_{t-1}$	-0.975^{***}	-1.149^{***}	2.717	1.634
	(0.211)	(0.221)	(4.965)	(5.513)
Mkt-rf × $Surprise_{t-1}$	5.027	5.149		
	(8.949)	(8.692)		
SMB × $Surprise_{t-1}$	−9.881	−10.40		
	(20.27)	(22.88)		
HML × $Surprise_{t-1}$	25.44	30.12		
	(25.35)	(26.35)		
UMD × $Surprise_{t-1}$	27.74^{*}	43.43^{***}		
	(14.71)	(14.35)		
Own $Surprise_{it}$ controls	Yes	Yes	Yes	Yes
R^2	0.0616	0.196	0.891	0.754
Observations	75,897	76,062	75,734	68,750

(*Continued*)

Table VII—*Continued*

Panel B: Potential Market Effects

	Return $[t-1, t+1]$ (1)	(2)	(3)	(4)
$Surprise_{t-1}$	-0.923^{***}		-0.921^{***}	-0.965^{***}
	(0.233)		(0.231)	(0.226)
$Market_{t-1}$	-0.0641^{*}	-0.0644^{*}	-0.0748^{*}	-0.0721^{*}
	(0.0369)	(0.0357)	(0.0420)	(0.0421)
$Surprise_{t-1}$ × $Market_{t-1}$			6.141	7.882
			(12.32)	(12.39)
$Market_t$				0.0457
				(0.0500)
$Surprise_{t-1}$ × $Market_t$				14.39
				(15.09)
Own $Surprise_{it}$ controls	Yes	Yes	Yes	Yes
R^2	0.0603	0.0591	0.0603	0.0606
Observations	75,897	75,897	75,897	75,897

Panel C: Limited Capital

	Excl quarters ret$< -.05$	VIX		Earnings arb capital	
Return $[t-1, t+1]$	(1)	(2)	(3)	(4)	(5)
$Surprise_{t-1}$	-1.060^{***}	-0.968^{***}	-1.004^{*}	-0.916^{***}	-0.855^{***}
	(0.248)	(0.252)	(0.596)	(0.250)	(0.294)
VIX		0.0000564	0.0000550		
		(0.0000724)	(0.0000778)		
$Surprise_{t-1}$ × VIX			0.00161		
			(0.0266)		
Arb capital				−0.231	−0.207
				(0.282)	(0.293)
$Surprise_{t-1}$ × Arb capital					−41.40
					(116.1)
Own $Surprise_{it}$ controls	Yes	Yes	Yes	Yes	Yes
R^2	0.0643	0.0646	0.0646	0.0602	0.0602
Observations	63,730	70,964	70,964	67,127	67,127

Read: The negative $Surprise_{t-1}$ effect remains after risk-factor interactions, liquidity proxies, market returns, VIX, and earnings-arbitrage-capital controls. In Panel C, the main coefficient remains around −0.86 to −1.06 across specifications.

Economic message: The contrast effect is not a disguised risk premium or a simple liquidity/capital-constraint effect. The baseline behavioral interpretation survives the main asset-pricing alternatives.

5.7 Tables VIII and IX: strategic timing and robustness

Table VIII

Strategic Timing of Earnings Announcements

This table tests whether the negative relation between return reactions and $Surprise_{t-1}$ is driven by changes in the scheduling of announcements. Δdate is the difference between the day of the current earnings announcement and the previous year's same-quarter earnings announcement (e.g., for a firm announcing on March 15, 2004 that previously announced on March 12, 2003, $\Delta\text{date} = 3$). All other variables and weights are as defined in Table II. Standard errors are clustered by date and reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Return $[t-1, t+1]$	
	(1)	(2)
$Surprise_{t-1} \times \text{abs}(\Delta \text{ date}) \leq 5$	-0.945^{***}	
	(0.240)	
$Surprise_{t-1} \times \text{abs}(\Delta \text{ date}) > 5$	-0.716	
	(0.753)	
$Surprise_{t-1} \times \Delta \text{ date} < -5$		0.797
		(0.982)
$Surprise_{t-1} \times \text{abs}(\Delta \text{ date}) \leq 5$		-0.945^{***}
		(0.240)
$Surprise_{t-1} \times \Delta \text{ date} > 5$		-1.213
		(0.924)
Own $Surprise_{it}$ controls	Yes	Yes
R^2	0.0608	0.0612
Observations	70,091	70,091

Table IX

Robustness to Different Measures of $\boldsymbol{Surprise_{t-1}}$

	Return $[t-1, t+1]$					
	(1)	(2)	(3)	(4)	(5)	(6)
Return surprise$_{t-1}$, VW mean	-0.0520^{***}	-0.0492^{**}				
	(0.0197)	(0.0247)				
$Surprise_{t-1}$, volume-weighted			-0.285^{**}			
			(0.130)			
$Surprise_{t-1}$, analyst-weighted				-0.915^{***}		
				(0.218)		
$Surprise_{t-1}$, no price scaling					-0.0310^{***}	
					(0.00667)	
$Surprise_{t-1}$, scaled *SD*						-0.486^{***}
						(0.130)
Own $Surprise_{it}$ controls	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0172	0.0254	0.0580	0.0600	0.0596	0.0600
Observations	136,056	74,897	73,472	75,897	75,923	75,897

Read: Table VIII shows the effect remains for firms whose announcement date is within five days of the previous year's same-quarter announcement date. Table IX shows robustness to return-based, volume-weighted, analyst-weighted, unscaled, and scaled-standard-deviation versions of $Surprise_{t-1}$.

Economic message: The result is not driven by managers strategically choosing announcement dates or by one particular way of measuring earnings surprise.

5.8 Table X: timing of contrast effects

Table X
Timing of Contrast Effects

Panel A: Further Lags and Leads

	Return $[t-3, t+1]$		Return $[t-1, t+3]$	
	(1)	(2)	(3)	(4)
$Surprise_{t-3}$	-0.223	-0.133		
	(0.227)	(0.266)		
$Surprise_{t-2}$	0.260	0.0664		
	(0.270)	(0.871)		
$Surprise_{t-1}$	-0.728^{***}	-1.198^{**}	-1.045^{***}	-0.778^{**}
	(0.247)	(0.480)	(0.279)	(0.314)
$Surprise_{t+1}$			0.0689	-0.143
			(0.393)	(0.494)
$Surprise_{t+2}$			-0.395	-0.263
			(0.379)	(0.681)
p-Value: $(t-3) = (t-1)$	0.117	0.0398		
p-Value: $(t-2) = (t-1)$	0.00894	0.216		
p-Value: $(t+1) = (t-1)$			0.0219	0.278
p-Value: $(t+2) = (t-1)$			0.172	0.487
Days	All	Th Fr	All	Mo Tu
Own $Surprise_{it}$ controls	Yes	Yes	Yes	Yes
R^2	0.0575	0.0741	0.0530	0.0577
Observations	75,844	29,376	75859	25408

(*Continued*)

Table X—*Continued*

Panel B: Same-Day Contrast Effects

Own announcement time:	Today	PM		AM		AM or PM
	(1)	(2)	(3)	(4)	(5)	(6)
Others' announcement time:	Today	AM	PM	AM	PM	Prev halfday
Others' surprise	-0.587^{*}	-1.263^{*}	0.426	-0.417	-0.444	-0.479
	(0.307)	(0.664)	(0.694)	(0.408)	(0.343)	(0.375)
Own $Surprise_{it}$ controls	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0558	0.0900	0.0805	0.0641	0.0586	0.0738
Observations	79,886	19,300	20,059	21,824	17,899	36,901

Panel C: Day of the Week

Own announcement:	Monday	Tuesday	Wednesday	Thursday	Friday	Tu-Th
	(1)	(2)	(3)	(4)	(5)	(6)
$Surprise_{t-1}$	-1.170^{**}	-0.583^{*}	-0.706	-1.005^{*}	-2.578^{***}	-0.686^{***}
	(0.460)	(0.298)	(0.623)	(0.541)	(0.896)	(0.248)
Own $Surprise_{it}$ controls	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0875	0.0702	0.0407	0.0735	0.104	0.0577
Observations	7,743	17,677	21,083	23,577	5,817	62,337

Panel D: Quarter of the Year

Quarter (Calendar Year)	Q1	Q2	Q3	Q4
	(1)	(2)	(3)	(4)
$Surprise_{t-1}$	-1.331^{***}	-0.857^{***}	-0.757	-0.848^{*}
	(0.483)	(0.330)	(0.739)	(0.456)
Own $Surprise_{it}$ controls	Yes	Yes	Yes	Yes
R^2	0.0513	0.0714	0.0717	0.0594
Observations	16,185	20,935	18,783	19,994

Read: Panel A shows that the previous trading day's surprise matters more than $t-2$, $t-3$, $t+1$, or $t+2$ surprises. Panel B shows within-day evidence: morning surprises negatively affect afternoon announcement returns. Panels C and D show the effect across weekdays and quarters.

Economic message: Recency is central. Investors contrast today's news with the most recent salient comparison signal.

5.9 Tables XI and XII: heterogeneity and comparison groups

Table XI
Heterogeneity

	Return $[t-1, t+1]$		
	(1)	(2)	(3)
$Surprise_{t-1}$ × size quintile 1	−0.527		
	(0.480)		
$Surprise_{t-1}$ × size quintile 2	-0.792^{*}		
	(0.475)		
$Surprise_{t-1}$ × size quintile 3	−0.324		
	(0.443)		
$Surprise_{t-1}$ × size quintile 4	0.179		
	(0.304)		
$Surprise_{t-1}$ × size quintile 5	-1.028^{***}		
	(0.255)		
$Surprise_{t-1}$ × (num analysts = 1)		0.0428	
		(0.607)	
$Surprise_{t-1}$ × (num analysts = 2)		-1.048^{**}	
		(0.508)	
$Surprise_{t-1}$ × (num analysts ≥ 3)		-1.020^{***}	
		(0.256)	
$Surprise_{t-1}$ × 1980s			−0.663
			(0.455)
$Surprise_{t-1}$ × 1990s			-1.024^{*}
			(0.545)
$Surprise_{t-1}$ × 2000s			−0.542
			(0.346)
$Surprise_{t-1}$ × 2010s			-1.001^{**}
			(0.462)
Own $Surprise_{it}$ controls	Yes	Yes	Yes
R^2	0.0604	0.0604	0.0631
Observations	75,897	75,897	75,897

Table XII
Industry Match

	Full Sample			Small Firms		Large Firms	
Return $[t+1, t-1]$	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Surprise_{t-1}$ same ind	-0.507^{***}	-0.421^{***}	-0.520^{***}	-0.746^{***}	-0.794^{***}	-0.497^{***}	-0.511^{***}
	(0.180)	(0.133)	(0.184)	(0.228)	(0.239)	(0.184)	(0.189)
$Surprise_{t-1}$ dif ind	-0.489^{***}	-0.171^{*}	-0.321^{*}	-0.440^{**}	−0.214	-0.487^{***}	-0.321^{*}
	(0.161)	(0.104)	(0.184)	(0.206)	(0.248)	(0.166)	(0.189)
Both $Surprise_{t-1}$ nonmissing	No	No	Yes	No	Yes	No	Yes
Regression weights	Value	Equal	Value	Value	Value	Value	Value
p-Value: same=dif	0.944	0.181	0.462	0.355	0.123	0.970	0.495
Own $Surprise_{it}$ controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0605	0.0682	0.0557	0.0833	0.0873	0.0595	0.0544
Observations	75,897	75,897	49,343	33,831	20,847	42,066	28,496

Read: Table XI shows strong effects for large firms, for firms covered by two or more analysts, and across several decades. Table XII separates same-industry and different-industry previous-day surprises. Same industry comparisons are especially important for small firms.

Economic message: Investors' comparison sets are not arbitrary. Salience and similarity matter: large firms are compared with other large firms, and smaller firms are more affected by same-industry comparison news.

6 Short summary

- The paper documents contrast effects in financial markets using earnings announcements.
- The key variable is the earnings surprise of large firms that announced on the previous trading day.
- The key outcome is the characteristic-adjusted return reaction to today's earnings announcement.
- The main result is negative: high previous-day surprises reduce today's return reaction, holding today's own surprise fixed.
- The effect is not driven by the interaction between yesterday's and today's surprises; it is mostly a directional shift in perception.
- The effect reverses over the next 50 trading days, especially from $t+26$ to $t+50$.
- A simple long-short strategy earns significant short-window abnormal returns.
- Information-transmission stories are weak because yesterday's surprise does not predict today's surprise after time controls, and prices do not move on $t-1$ as if the information is relevant.
- Risk, liquidity, market-wide capital constraints, strategic scheduling, and alternative surprise measures do not explain the result.
- The effect is strongest when the previous signal is recent and salient, and comparison groups seem shaped by firm size and industry similarity.

7 Conclusion

7.1 Core conclusion

Hartzmark and Shue show that investors evaluate earnings news in relative terms. A firm's earnings announcement receives a lower return reaction if it follows good earnings news from large firms and a higher return reaction if it follows bad earnings news.

7.2 Economic intuition

The market does not simply ask: "How good is this firm's news?" It also asks, implicitly and mistakenly: "How good does this news look relative to what I just saw?" This relative evaluation is psychologically natural but economically irrelevant.

7.3 Takeaway

Contrast effects can generate mispricing in sophisticated markets. Attention and perception, not only information and incentives, are priced inputs in financial markets.